
Design Procedures for Fiber Reinforced Concrete Applications in New and Repair of hydraulic Structures

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**USCID 11th International Conference on Irrigation and Drainage
October 15-19, 2018**

FRC - Background

- high strength and ductile concrete has been made possible by advances in:
 - particle packing, aggregate gradation , superplasticizers, rheology
 - increased quality control
 - Innovative fibers, testing and specification methods
 - Chemical admixtures low water-to-cementing materials ratio.
- Ductility and crack width control that reduces liquid ingress, significantly enhancing durability



Problem Statement

- How do we use the existing knowledge on concrete material for better construction procedures and specifications for implementing the concrete materials?
- How do we capture the economic benefits of FRC in terms of speed of construction, reduced labor, better quality concrete?
- Cost-effective specification procedures for use in cast in place, precast, and shotcrete concrete applications.



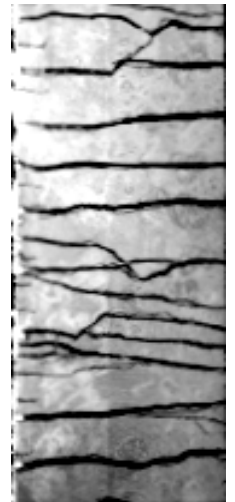
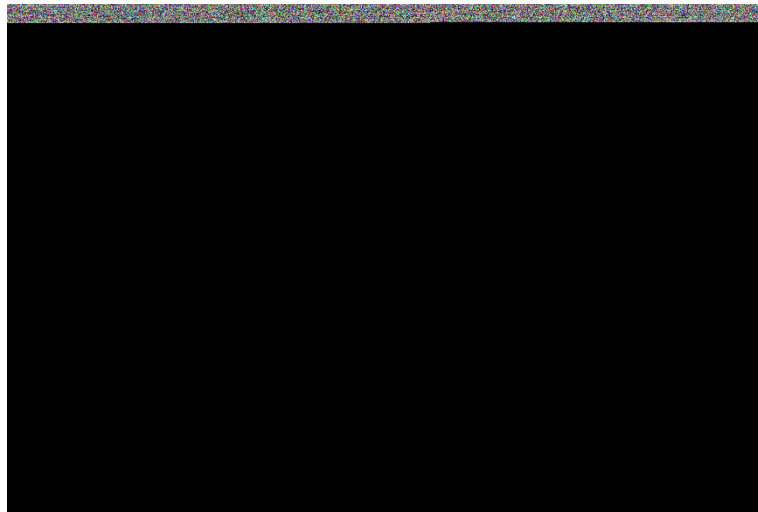
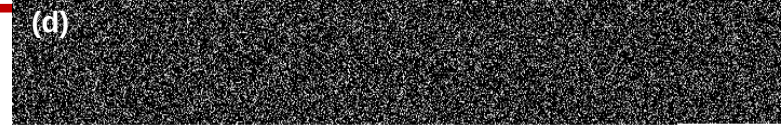
Objectives

- Existing procedures and standards on proportioning FRC mixtures and the fiber specific characteristics affecting the overall performance.
- Develop typical FRC mixtures with locally available materials
- Recommended Test methods and Specifications for qualification of mixtures for various applications
- Optimize the Structural and material design procedures based on performance criteria for strength, ductility, crack width and durability.
- Availability of specifications and testing guidelines for:
 - Mechanical: strength, ductility, impact, repair, volume changes and crack resistance
 - Durability: corrosion resistance, resistance to chloride ion ingress and freeze and thaw, shrinkage, curling, cracking.

Materials characteristics

- Ductility
- Toughening
- Improved tensile strength
- Increase level of energy absorption
- Fatigue life, impact/explosive loading
- Seismic resistance
- Steel work, labor, construction time.
- Corrosion damage
- Long-term repair and maintenance.

(d)



- Criteria for fibers in structural applications
- Criteria for repair of existing structures and canals
- Criteria for fibers in shotcrete
- Development of performance based specifications
- Design opportunities:
 - Ductility, durability, crack width, stiffness, cracked section modulus, Shear
 - Hybrid approach of combining reinforcement and fibers for sustainability
 - Minimum reinforcement requirements.
- Five major documents developed by the ACI 544-D committee
 - ACI 544-9R Mechanical testing of FRC
 - ACI 544-8R Tensile Design Properties from flexural tests, backcalculation procedures
 - ACI 544-4R Design
 - ACI 544-6R Elevated slabs (2015)
 - ACI 544-7R Tunnel lining

ACI related International Committee Reports

ACI 544.5R-10 “Physical Properties and Durability of Fiber-Reinforced Concrete,” Report, ACI Committee, p. 31, (2010).

ACI 544.6R-15 Report on Design and Construction of Steel Fiber-Reinforced Concrete Elevated Slabs (2015)

ACI 544.7R-16 Report on Design and Construction of Fiber-Reinforced Precast Concrete Tunnel Segments (2016)

ACI 544.8R-16: Report on Indirect Method to Obtain Stress-Strain Response of Fiber-Reinforced Concrete (FRC), ACI Committee 544 ACI 544.8R (2016)

ACI 544.9R-17: Report on Measuring Mechanical Properties of Hardened Fiber-Reinforced Concrete, (2017)

https://www.concrete.org/Portals/0/Files/PDF/Previews/544.9R-17_preview.pdf

ACI 544.10R-17: Report on Measuring Properties of Fresh Fiber-Reinforced Concrete, (2018)

ACI 544-4R ACI 544.4R-18: Report on Structural Design with Fiber-Reinforced Concrete (2018)

Report on Design and Construction of Fiber-Reinforced Precast Concrete Tunnel Segments

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V. Nasti is acknowledged as a significant contributor to this report. Special acknowledgements to M. Invernizzi, W. Bergeson, and S. Gholizadeh-Leonardi for their contributions to this report.

ally reinforced concrete segments such as saving cost and reducing production time while developing a more robust product with improved handling and long-term durability. Specific guidance on the design of fiber-reinforced precast concrete tunnel segments is needed for this emerging technology. This document offers general information on the history of FRC precast segments from tunneling projects throughout the world; a procedure for structural analysis and design based on governing load cases; and a description of the material parameters, tests, and analyses required to complete the design. The proposed guidelines are based on the knowledge gained from experimental research, analytical work, and the experience gained on numerous FRC precast tunnel projects.

Keywords: crack widths; earth pressure; fibers; fiber-reinforced concrete; grout pressure; hydrostatic pressure; lining; precast segment; stripping; surcharge load; thrust jack forces; tunnel.

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ACI 544.7R-16 was adopted and published January 2016.

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Report on Design and Construction of Steel Fiber-Reinforced Concrete Elevated Slabs

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Construction of slabs in areas with weak soil conditions has commonly used pile-supported slab structural design so that the adverse effects of soil-structure interaction in terms of differential settlement, cracking, or long-term serviceability problems are avoided. In this application, the construction of slabs on closely spaced pile caps (typical span-depth ratios between 8 and 30) is referred to as elevated ground slabs (EGSs). These slabs may be subjected to moderately high loading, such as concentrated point loading of up to 44 kip (150 kN) and uniformly distributed loadings of 1000 lb/ft² (50 kN/m²). The dynamic loadings may be due to moving loads such as forklifts, travel lifts, and other

material handling equipment. Fiber-reinforced concrete (FRC) has been successfully used to address the structural design of these slabs. Based on the knowledge gained, the area has been extended to a construction practice for slabs supported by columns as well. Applications are further extended to multi-story building applications. This report addresses the methodology for analysis, design, and construction of steel FRC (SFRC) slabs supported on piles or columns (also called elevated SFRC [E-SFRC]). Sections of the report address the history, practice, applications, material testing, full-scale testing, and certifications. By compiling the practice and knowledge in the analysis design with FRC materials, the steps in the design approach based on ultimate strength approach using two-way slab mechanisms are presented. The behavior of a two-way system may not require the flexural strength of conventional reinforced concrete (RC) because of redistribution, redundancy, and failure mechanisms.

Methods of construction, curing, and full-scale testing of slabs are also presented. A high dosage of deformed steel fibers (85 to 170

ACI 544.6R-15 was adopted and published September 2015.

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Fiber reinforcement has emerged as an alternative to traditional reinforcing bars and welded wire mesh reinforcement for precast concrete tunnel segments. Due to significantly improved post-cracking behavior and crack control characteristics, fiber-reinforced concrete (FRC) segments offer advantages over traditional

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Report on Measuring Mechanical Properties of Hardened Fiber-Reinforced Concrete

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The committee would like to thank H. Aoude and F. Vossoughi for their contributions to this report.

This report provides a synopsis of the existing testing methodologies for the determination of mechanical properties of hardened fiber-reinforced concrete (FRC). This report applies to the mechanical properties of conventionally mixed and placed FRC, including fiber-reinforced self-consolidating concrete (FRSC), or fiber-reinforced shotcrete (FRS) using steel, glass, polymeric, and natural fibers.

The objective is to enable manufacturers to characterize the mechanical properties of hardened FRC and encourage researchers and testing laboratories to adopt common and unified test methods to build a meaningful database of mechanical properties of hard-

ened FRC materials and products. Test results from the test procedures used in this report are not intended for the design of FRC structures, but to gain a better understanding of factors influencing the determination of their mechanical properties and of FRCs and FRC products.

Keywords: compressive strength; fiber pullout; fiber-reinforced concrete; flexural fatigue resistance; flexural strength; impact resistance; multiaxial behavior; shear and torsion; tensile strength; toughness.

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ACI 544.9R-17 was adopted and published January 2017.

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Report on Indirect Method to Obtain Stress-Strain Response of Fiber-Reinforced Concrete (FRC)

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Development of proper design procedures for fiber-reinforced concrete (FRC) materials requires use of material tensile and compressive stress strains that reflect the contribution of fibers to the post-cracking behavior. While uniaxial tension tests provide the most fundamental material properties, conducting closed-loop tension tests are difficult to accomplish; therefore, methods based on indirect measurement of tensile properties using flexural tests are typically used.

This report presents the methodologies that are used for data reduction and presentation of the flexural test results in terms of an equivalent tensile stress-strain response for FRC materials. Existing methods for estimating uniaxial tensile stress-strain response of strain-softening and hardening FRCs from flexural beam-test data are introduced. Different approaches applied to beam tests based on elastic equivalent, curve fitting, or back-cal-

ulation of flexural data are introduced. These are divided into two general categories: elastic equivalent approach or inverse analysis method. In the elastic equivalent approach, a summary of available test methods by various code agencies are presented.

Using back-calculation methods, tools based on the finite element method and analytical closed-form solutions are presented. An approach is presented that uses closed-form moment-curvature relationships and obtains load-deflection responses for a beam of three- or four-point loading. The method is used to obtain equivalent parametric tensile stress and strain relationships for a variety of FRC materials. The methods are compared against the available residual strength and also elastically equivalent residual strengths obtained by different specimen geometries.

Results for a range of FRC materials studied show the back-calculated post-peak residual tensile strength is approximately 39 to 37 percent of the elastically equivalent flexural residual strength for specimens with different fiber types and volume fractions.

Keywords: fiber-reinforced concrete; inverse analysis; tensile stress-strain diagram

ACI 544.8R-16 was adopted and published March 2016.

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FRC Engineering Applications

Ductility, Durability, Sustainability

- The type and volume fraction of fibers affect the level of energy absorption
- increased energy absorption, fatigue life, impact/explosive loading conditions, and seismic resistance
- Pavements/slabs, Precast components, Shotcrete



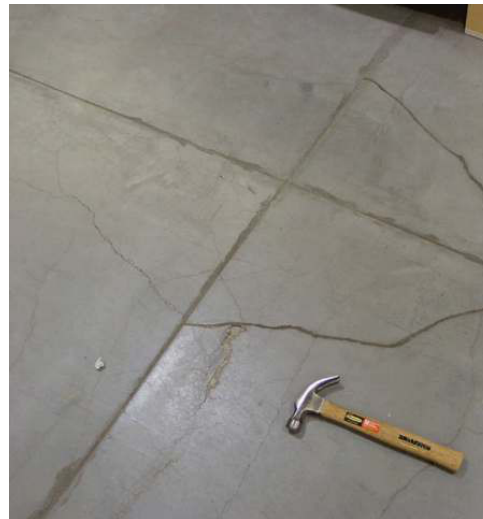
Shrinkage Induced Transverse Cracking

- Reduce load carrying capacity
- Accelerate deterioration
- Increase maintenance costs
- Reduce service life
- Plastic Shrinkage: Water evaporation at early age
- Drying Shrinkage: Loss of excessive water from hardened concrete



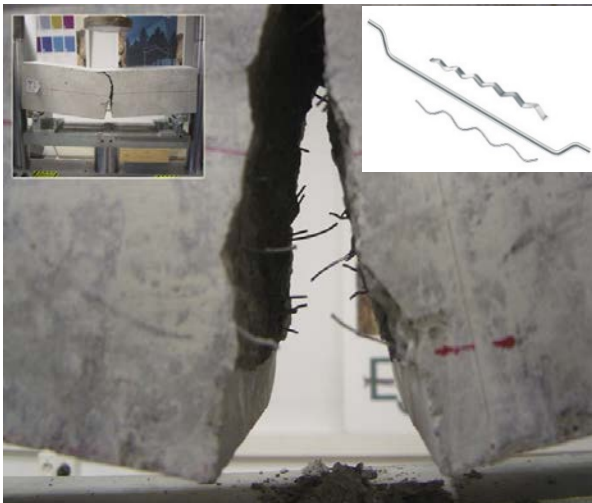
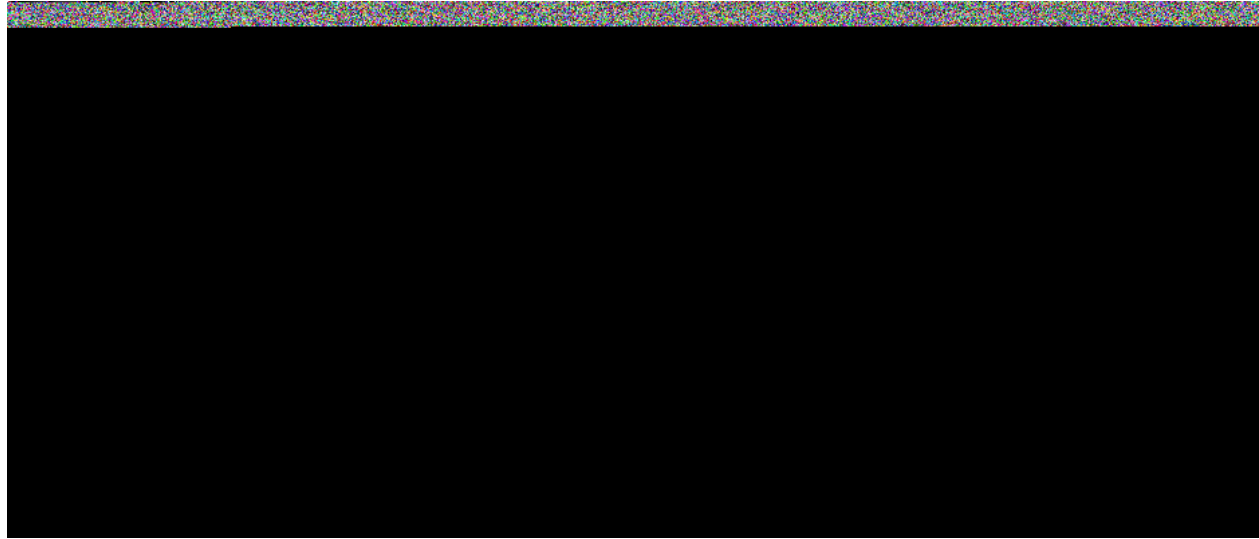
Joint failure

- Intended to control the cracks
- Presence of joints and corners aggravates the potential cracking as well as long term durability
- Joint edge chipping, uplift, and corner flexural cracks
- Potentially induced due to curling of the slab
- Point load such as truck wheel



Use of fibers in joint-free slab

- Crack bridging
- Load transfer
- Distributed fine cracks
- Non-critical crack pattern



Hybrid Reinforcement

Alternatives:

- 1- Conventional Steel Bars
- 2- Fiber Reinforcement
- 3- Hybrid Design

Issues with conventional reinforcement

- Significant **space**, **time** & **labor** needed to assemble cages & place reinforcing bars
- Spalling/Bursting **cracks** at segment joints/corners because of no reinforcement at concrete cover

Advantages of fiber reinforcement

- Improved precast production efficiency
- **Reduce** spalling or bursting **cracks** of concrete cover at vulnerable edges and corners
- Crack control, improved **serviceability** & **durability**
- High strength, ductility & robustness against **unintentional impact loads**



Segments near portal of Alaskan Way Viaduct/SR99 (Dia. 17.5m)– **Reinforcement Ratio > 120 kg/m³**
(Largest TBM tunnel in the world which is heavily reinforced but large cracks are observable)

Excess rebar specified can be replaced using a hybrid design approach in many applications

- Eliminate a substantial amount of reinforcing bars by using a highly flowable, steel /polymeric fiber-reinforced concrete
- The fibers are introduced during the concrete mixing process.



Canal lining, WWF, or rebar replacement



Traditional #5 rebar layout

Photo Courtesy: Rick Shelly, Pulice Construction

Fiber Reinforced Concrete Mix

Photo Courtesy: Pima-Maricopa Irrigation Project, Sacaton, Arizona

Safety, mobilization, and Cost savings due to reduced section sizes

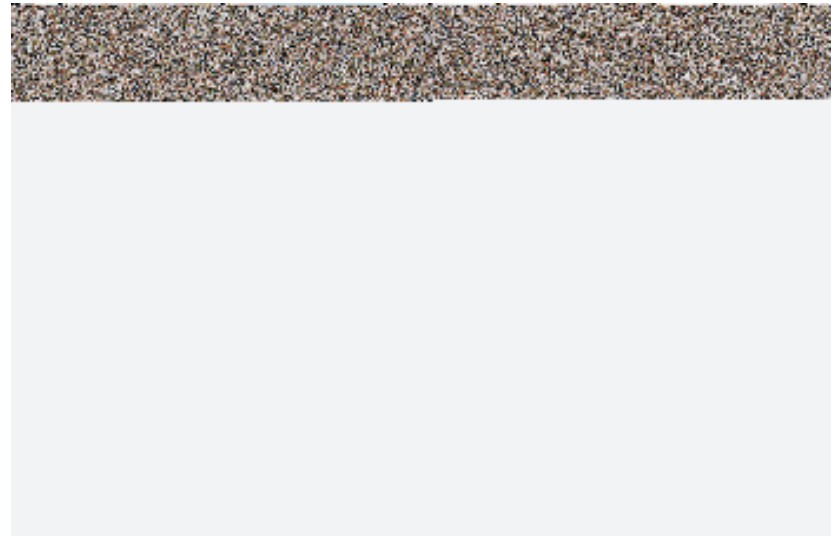


Tunnel Lining

- Fiber-reinforced concrete can be used in tunnel lining segments.
- Macro steel fibers and macro synthetic fibers are mainly used for this type of application.
- Using fibers can reduce or even eliminate the steel rebar reinforcement which results in faster and cheaper production.



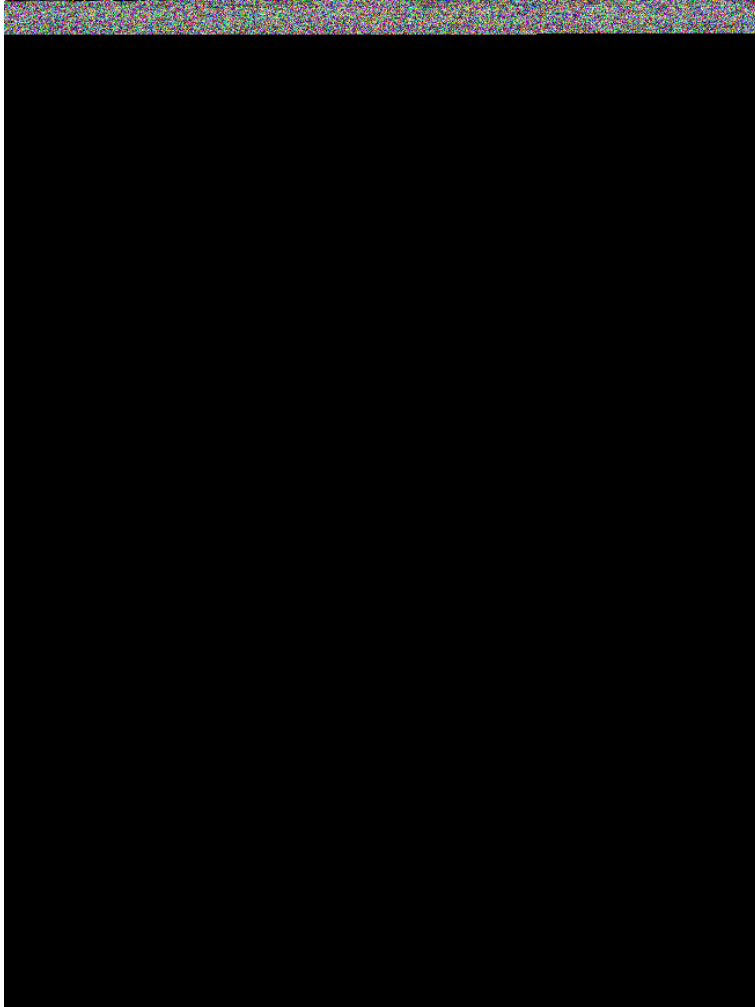
M. Moccichino et al., 2010 world
tunnel conference, Italy



BEKAERT

Structural Design with FRC Materials: testing, modeling, analysis and Design

Design with Strain Softening materials

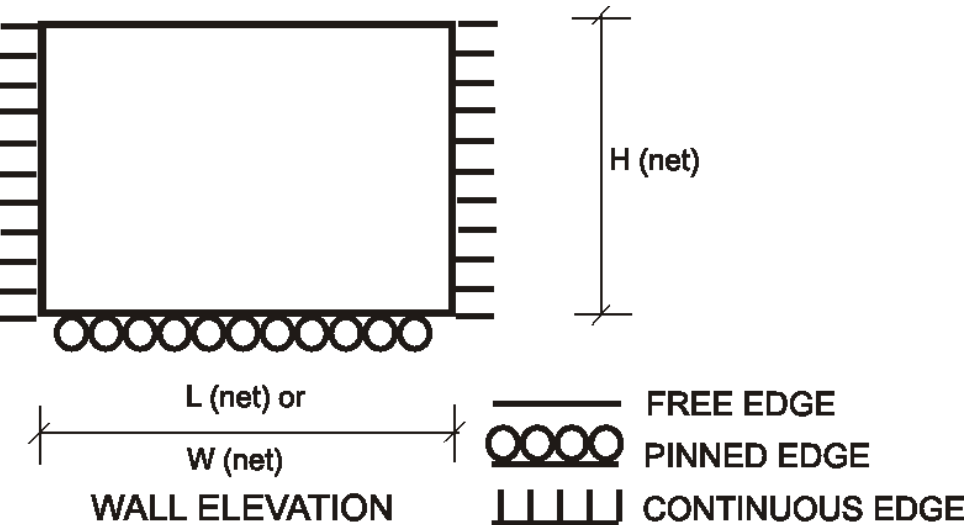


Precast panels

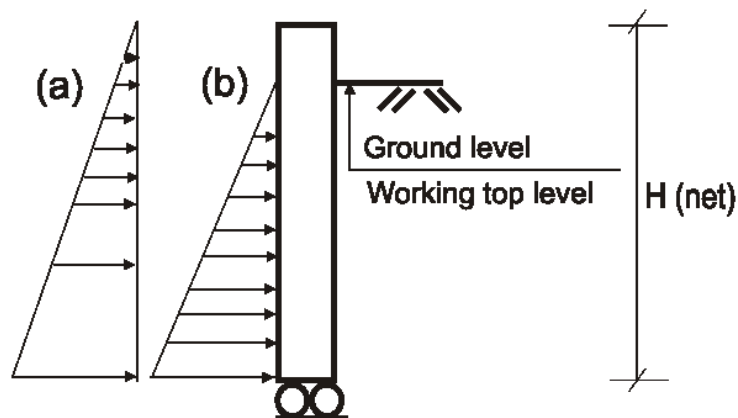


Shotcrete applications

Analysis of Precast Wall Panels



- Assume continuous wall, pin connection at the bottom and free at the top
- Lateral water pressure in ultimate and serviceability limit states

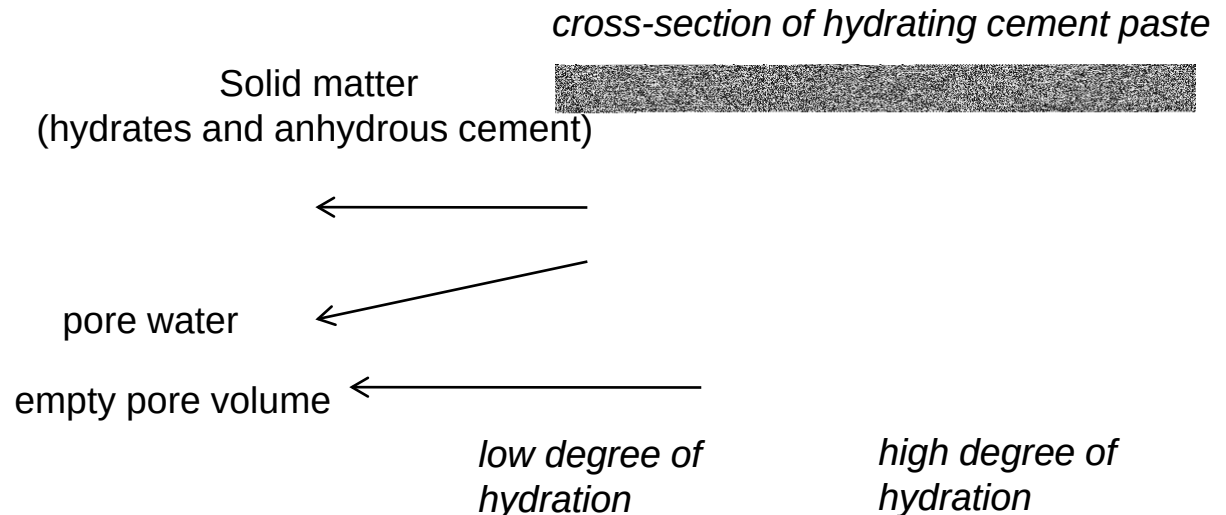
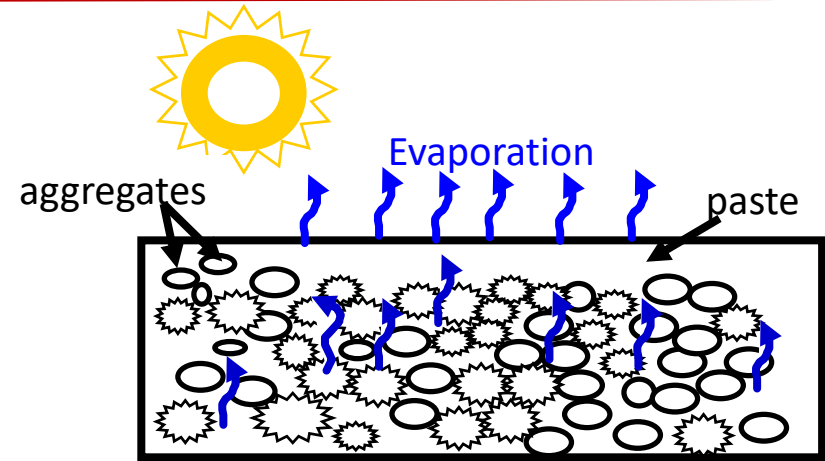


WALL SECTION

- (a) water distribution in Uls
(b) water distribution in Sls

Shrinkage as a Volumetric Change

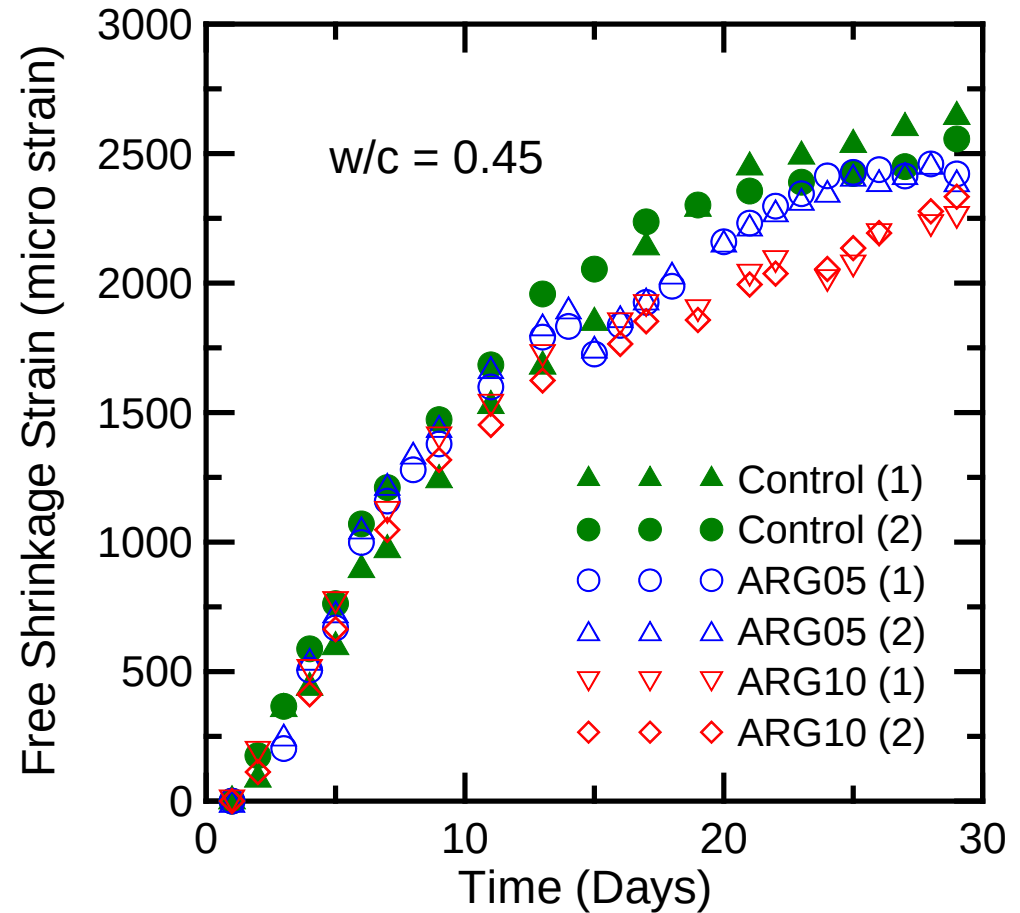
- Drying and plastic shrinkage
(due to water loss by evaporation)
- Autogenous/self-desiccation shrinkage
(due to cement hydration)
 - Thermal shrinkage (due to decrease in temperature)
- Carbonation shrinkage
(reaction of hydrated cement w/carbon dioxide)



Free shrinkage test for hardened concrete

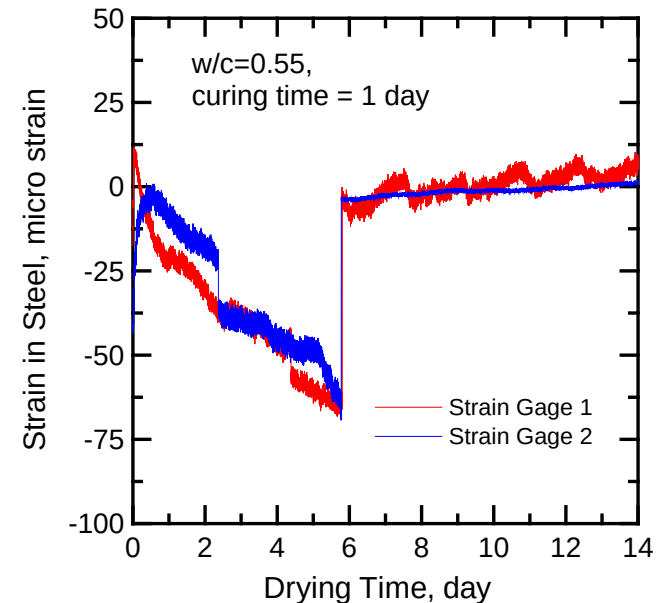
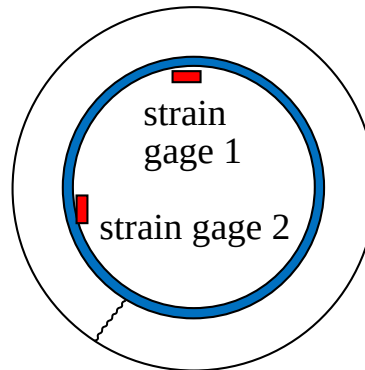
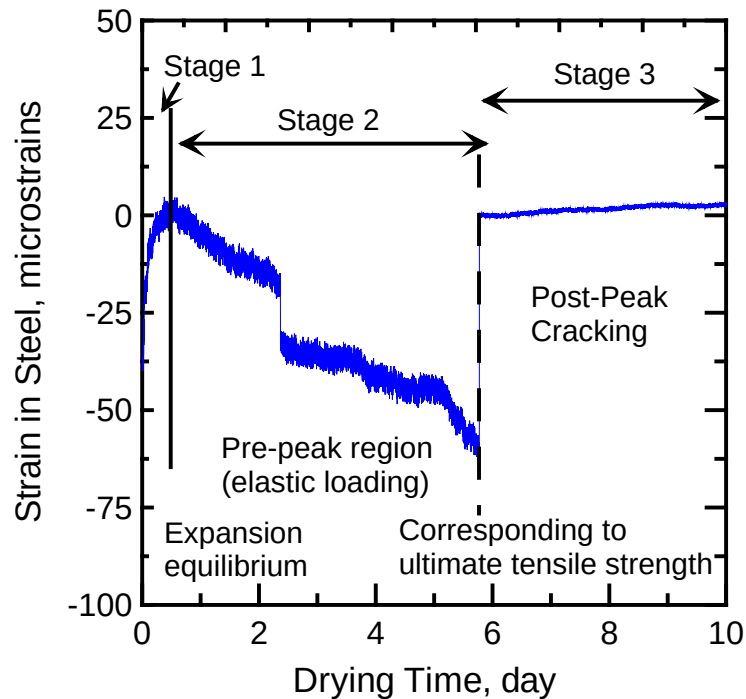


- Tests are performed according to ASTM C157



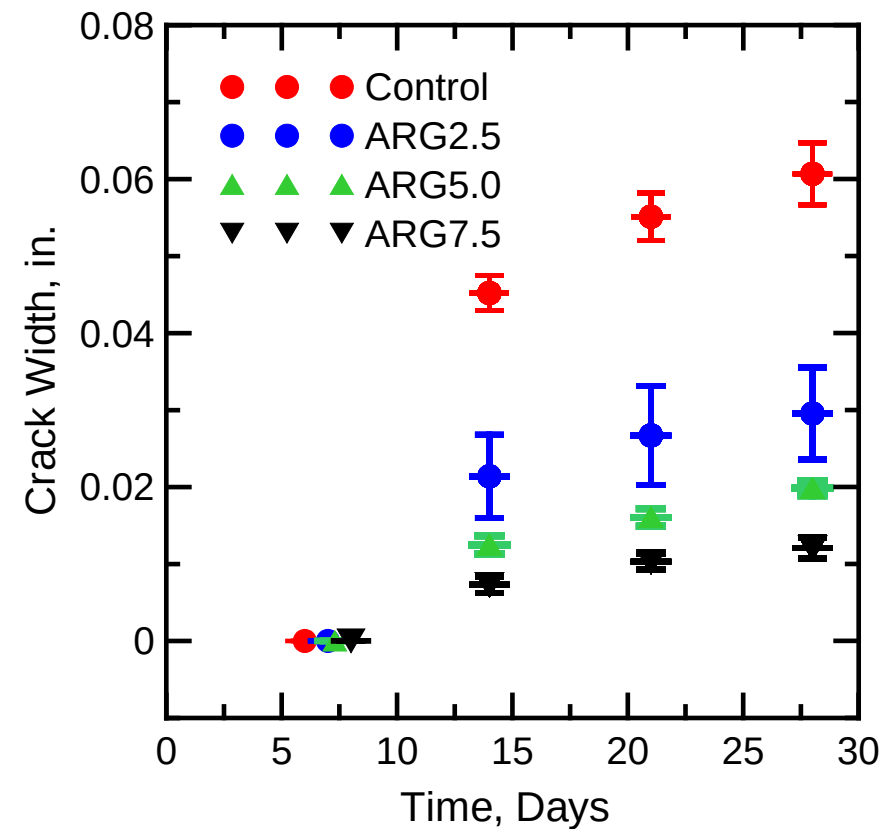
1D Shrinkage Cracking Ring Test AASHTO PP34-99

- Curing: 24 h under plastic sheets
- Relative humidity: $30\% \pm 1\%$
- Temperature: $40 \pm 1^\circ\text{C}$

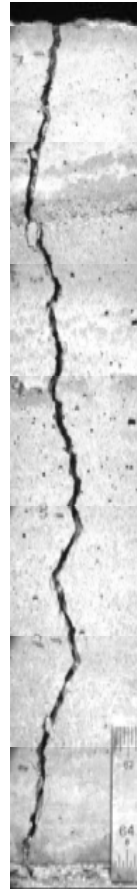
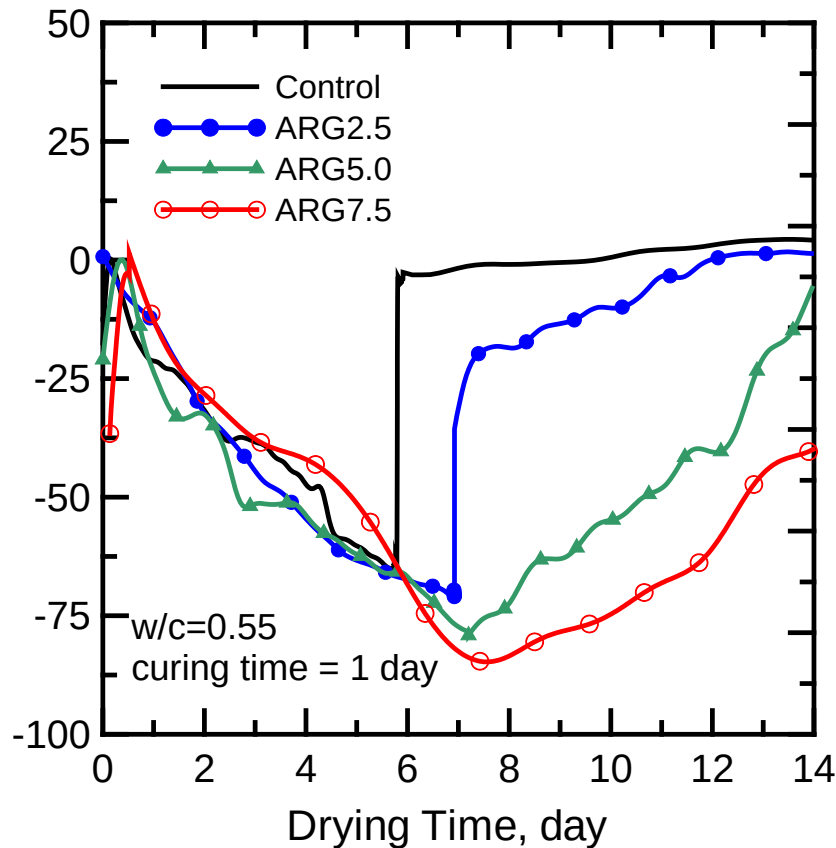


A typical result of a strain gage attached to steel ring

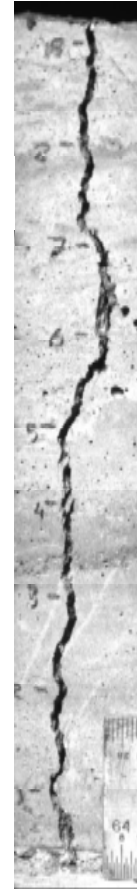
Image Analysis of 1D Restrained Shrinkage Cracks



Effect of fiber addition on strain in the steel rings



Average crack width: 1.15 mm
Standard deviation: 0.0787 mm

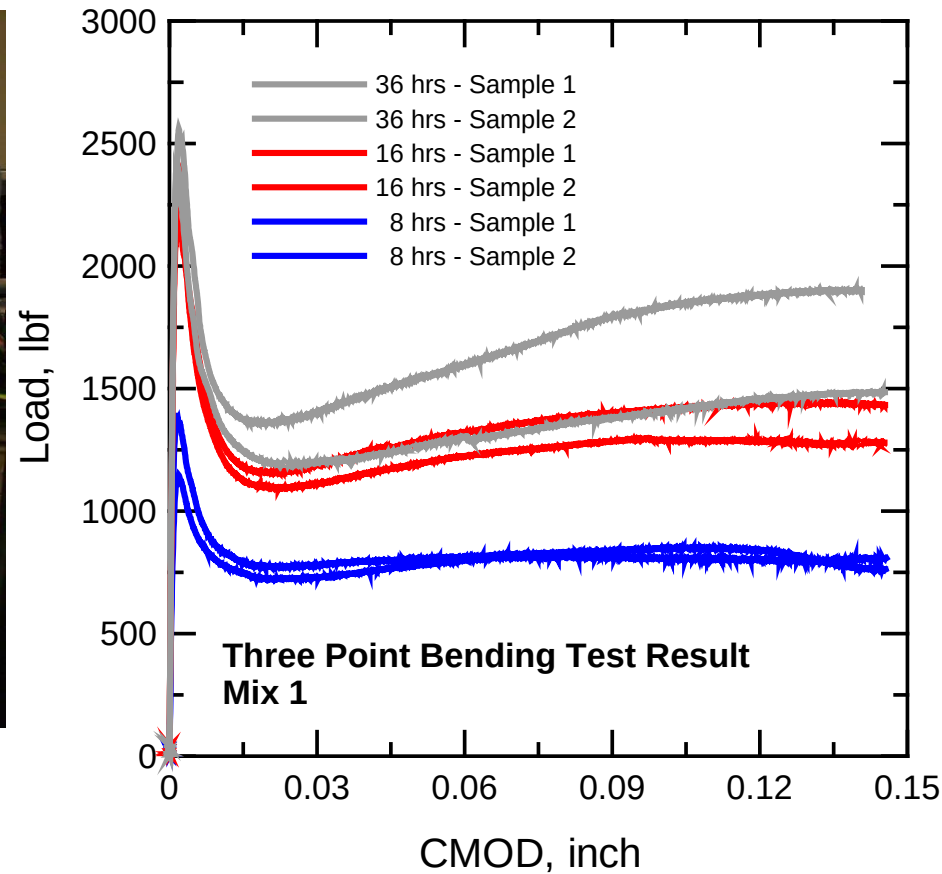


Average crack width: 0.341 mm
Standard deviation: 0.048 mm



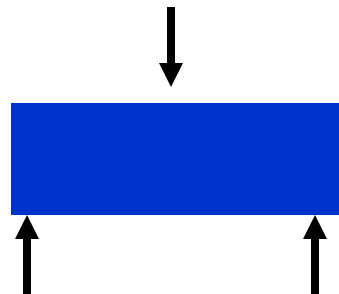
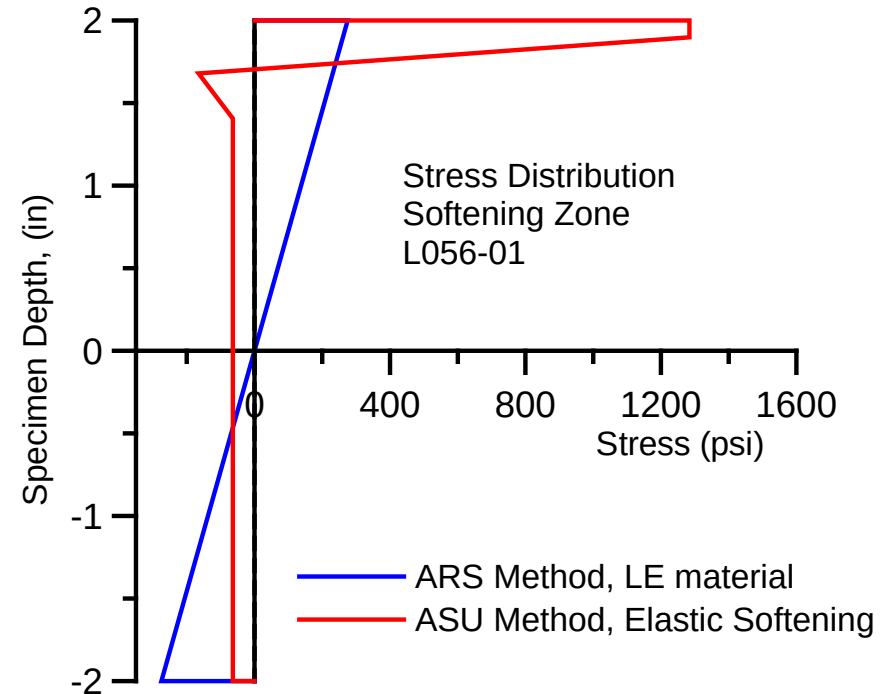
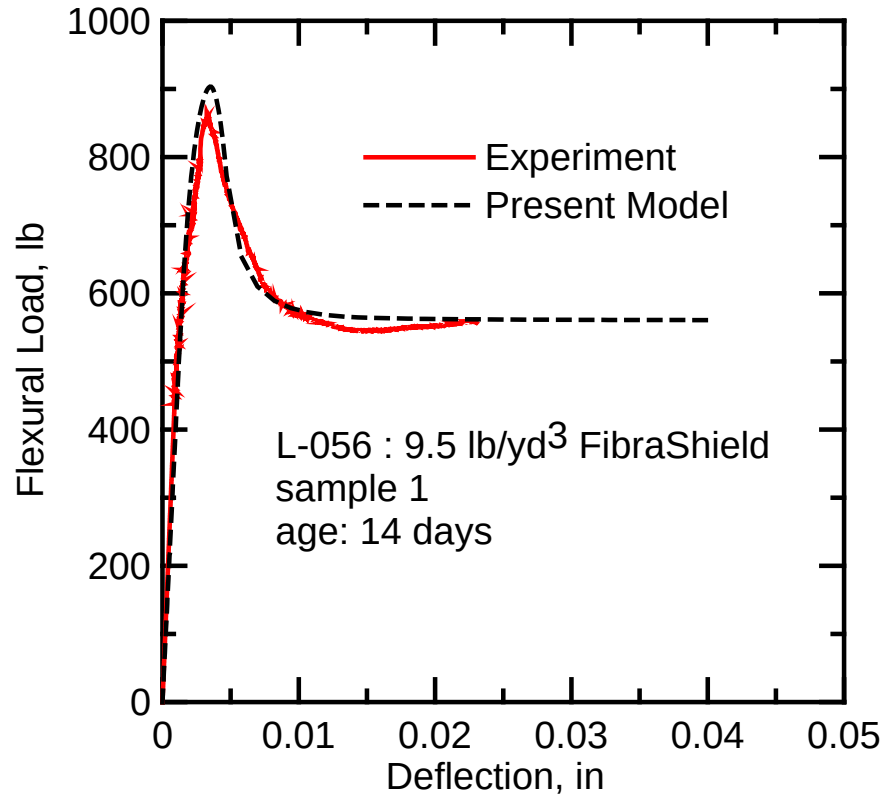
1. Extension of time to cracking
2. Post cracking ductility and strain distribution

Consideration of Residual strength Effect



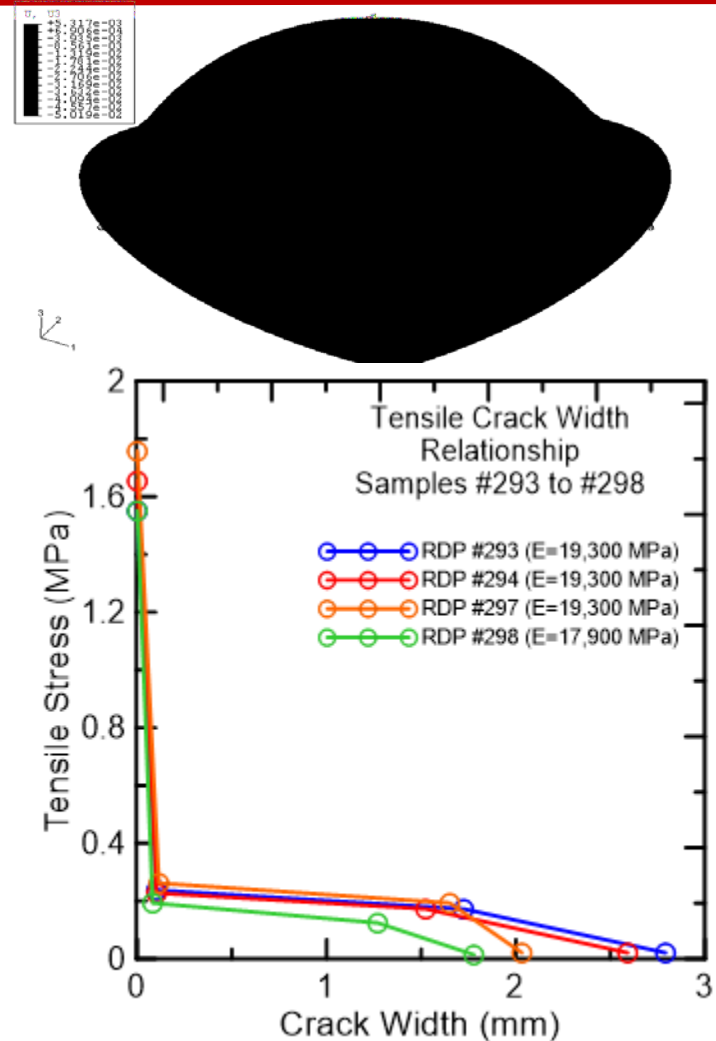
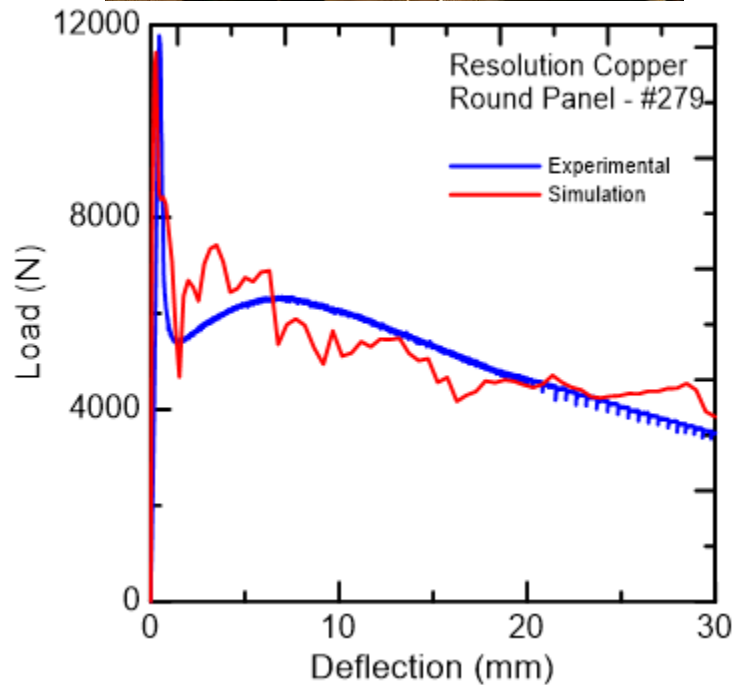
10-12 lbs/yd³ of macro fibers

Stress distribution for a cracked material with Strain-Softening FRC



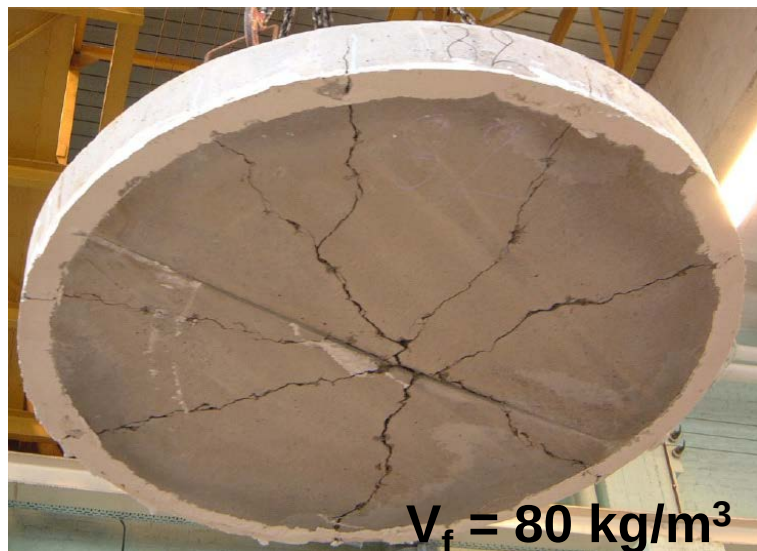
Shotcrete Applications

ASTM C1550-Round Panel 3P-support specimen

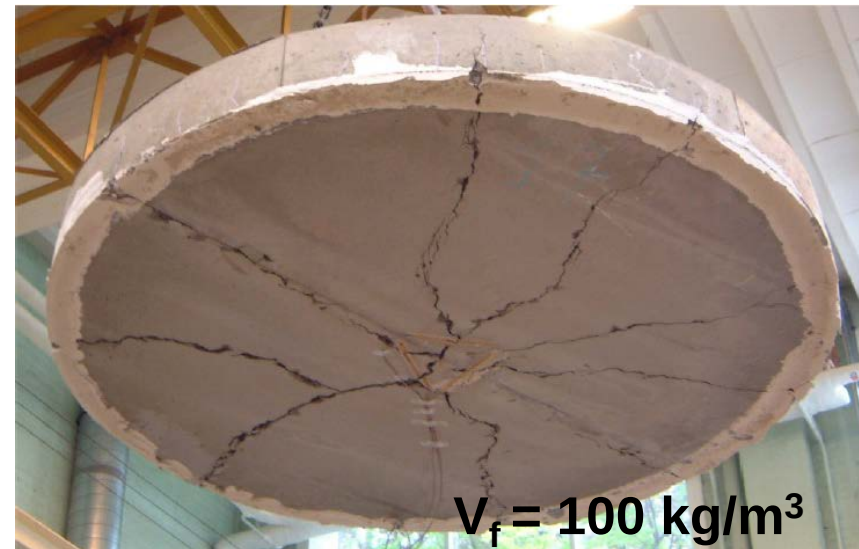


Round panel tests for evaluation of SFRC

- Test setup
 - displacement controlled, continuous support, center point load
- Dimensions
 - $D, t = 1500, 150$ mm
 - stroke diameter = 150 mm

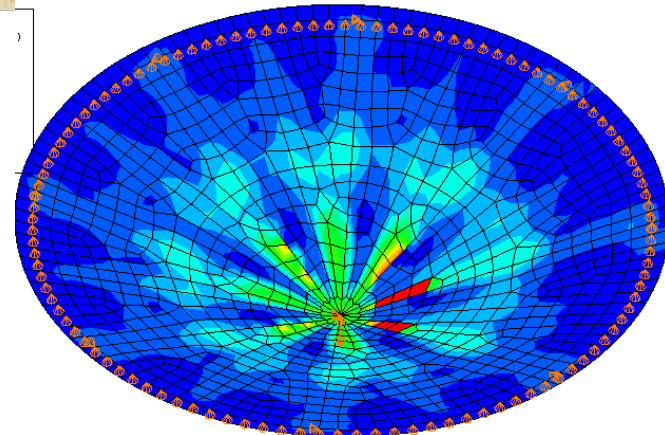
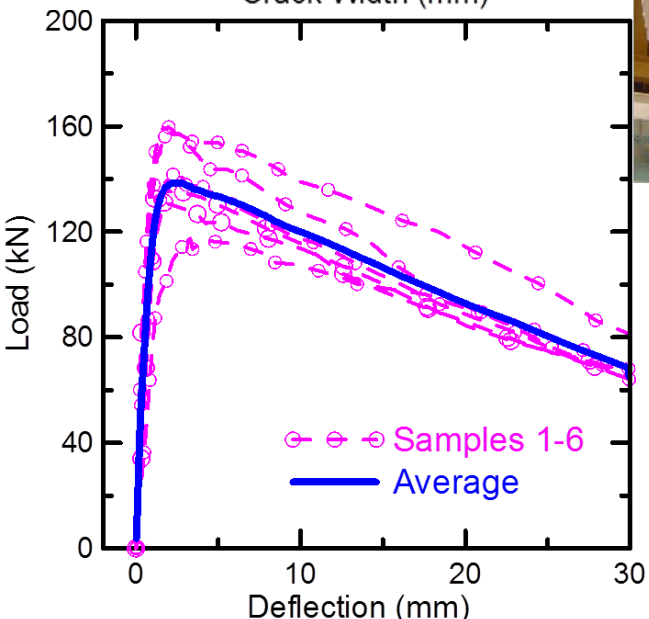
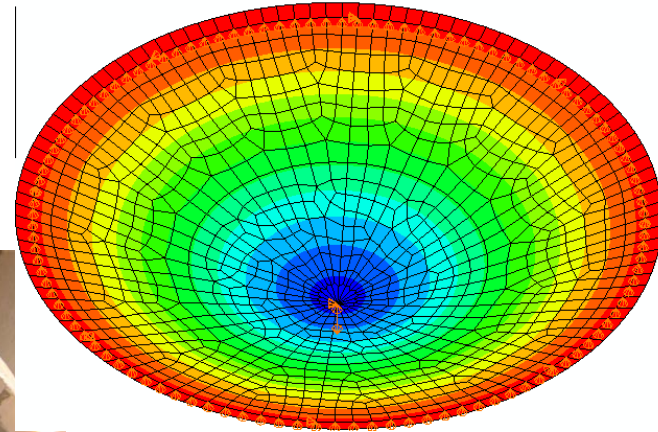
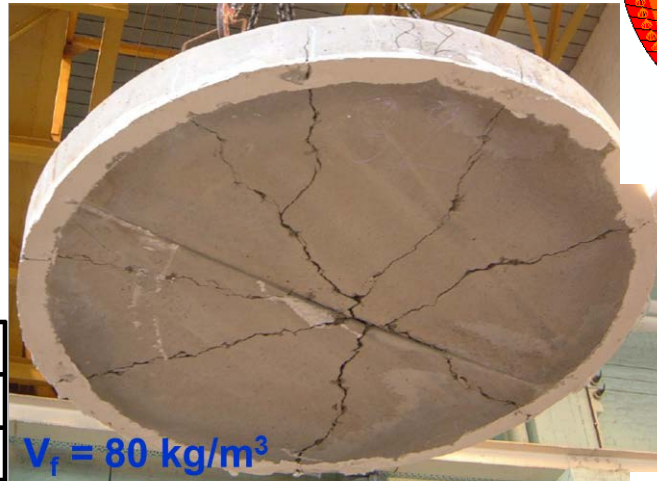
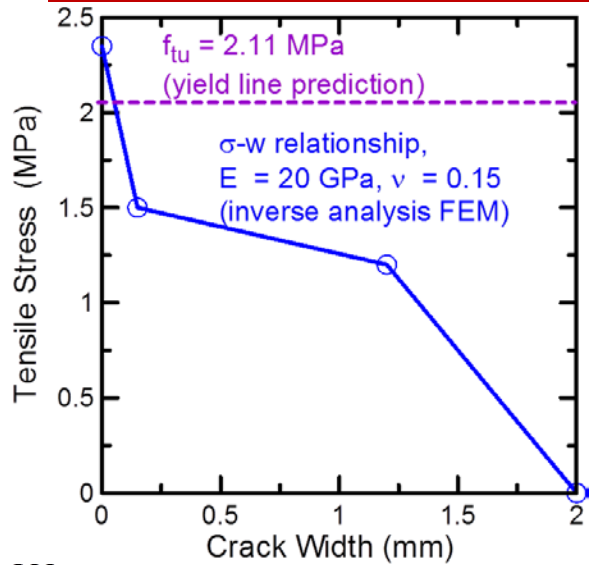


$V_f = 80 \text{ kg/m}^3$

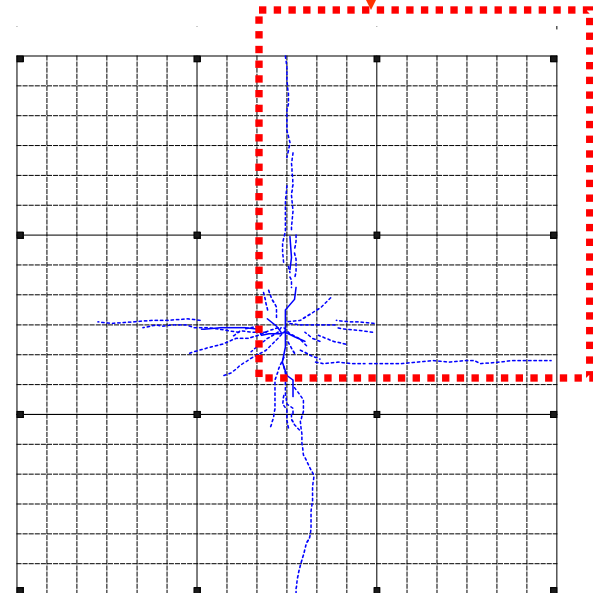
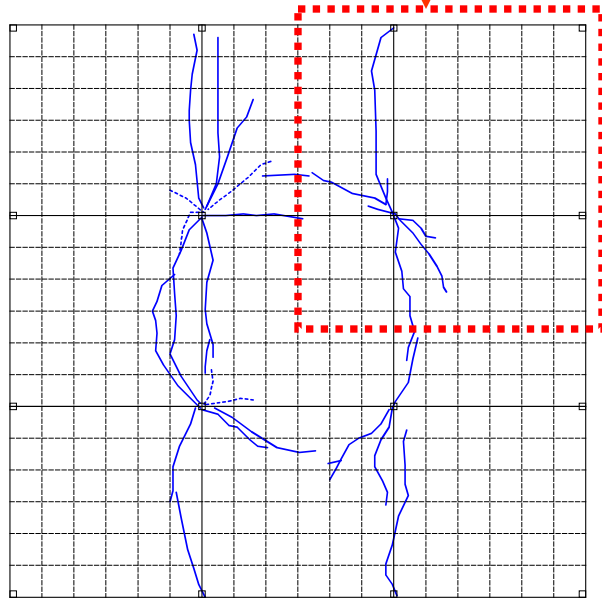
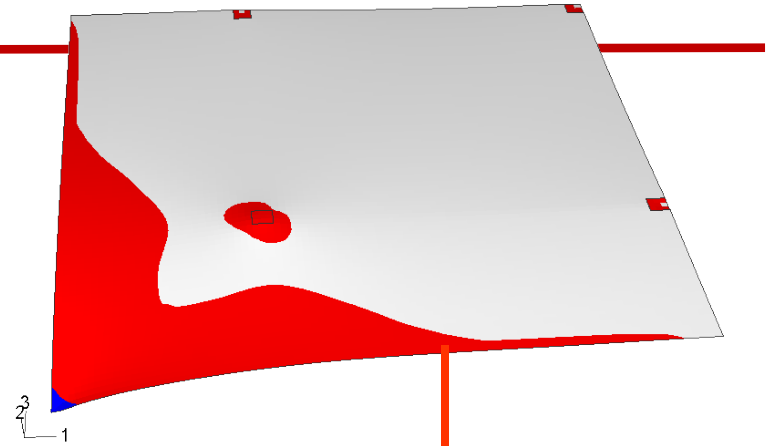
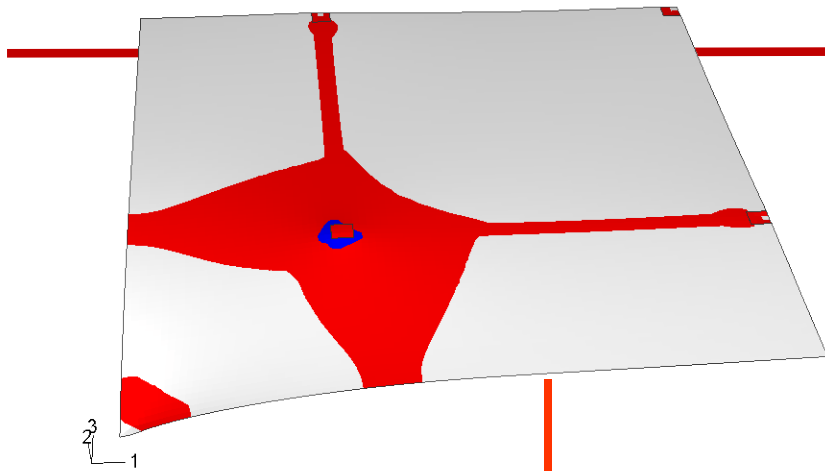


$V_f = 100 \text{ kg/m}^3$

Round Panel Continuous Support Specimens



Modeling of Failure Mechanisms



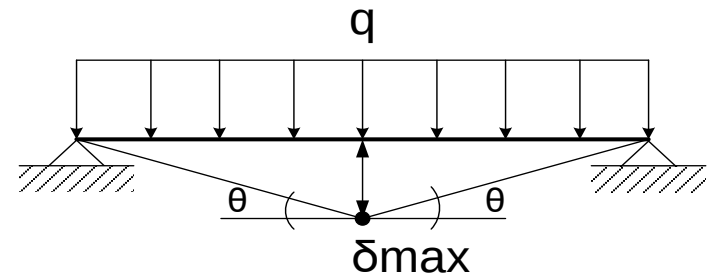
Plastic analysis approach

Distributed load on a simply supported square slab.

The work equations are derived as:

$$W_{int} = W_{ext}$$

Where the resultant N_R and rotation θ (from figure 9a) are:



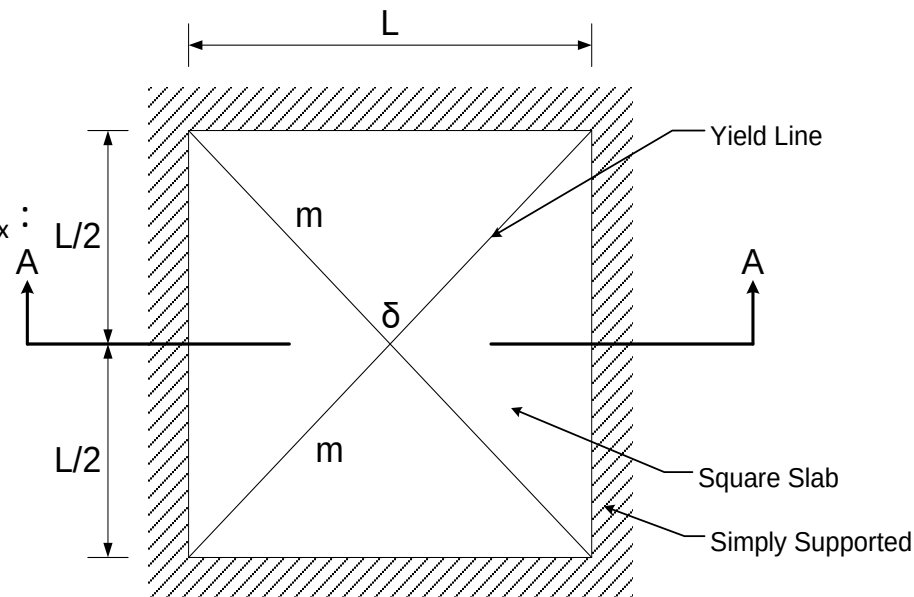
$$\sum (N_R \times \delta) = \sum (M \times L \times \theta)$$

$$N_R = q \times \left(\frac{L}{2}\right) \times \left(\frac{L}{2}\right) = \frac{qL^2}{4} \quad \theta = \frac{2\delta_{max}}{L}$$

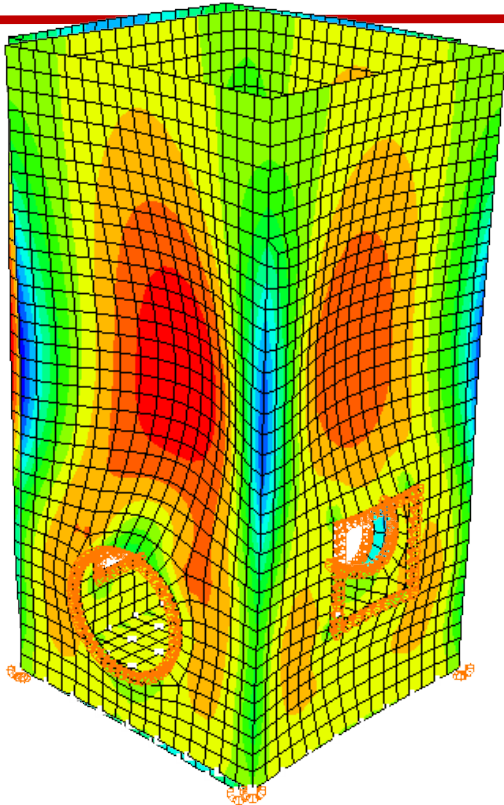
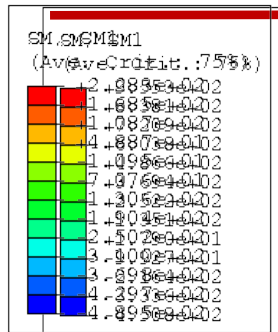
For the four segments with an N_R acting at $1/3$ of δ_{max} :

$$4 \times \left(\frac{qL^2}{4}\right) \times \left(\frac{\delta_{max}}{3}\right) = 4 \times (M) \times (L) \times \left(\frac{\delta_{max}}{\frac{L}{2}}\right)$$

$$M_p = \frac{q_{ult} L^2}{24}$$



Analysis, Design and Installation of precast water tank panels

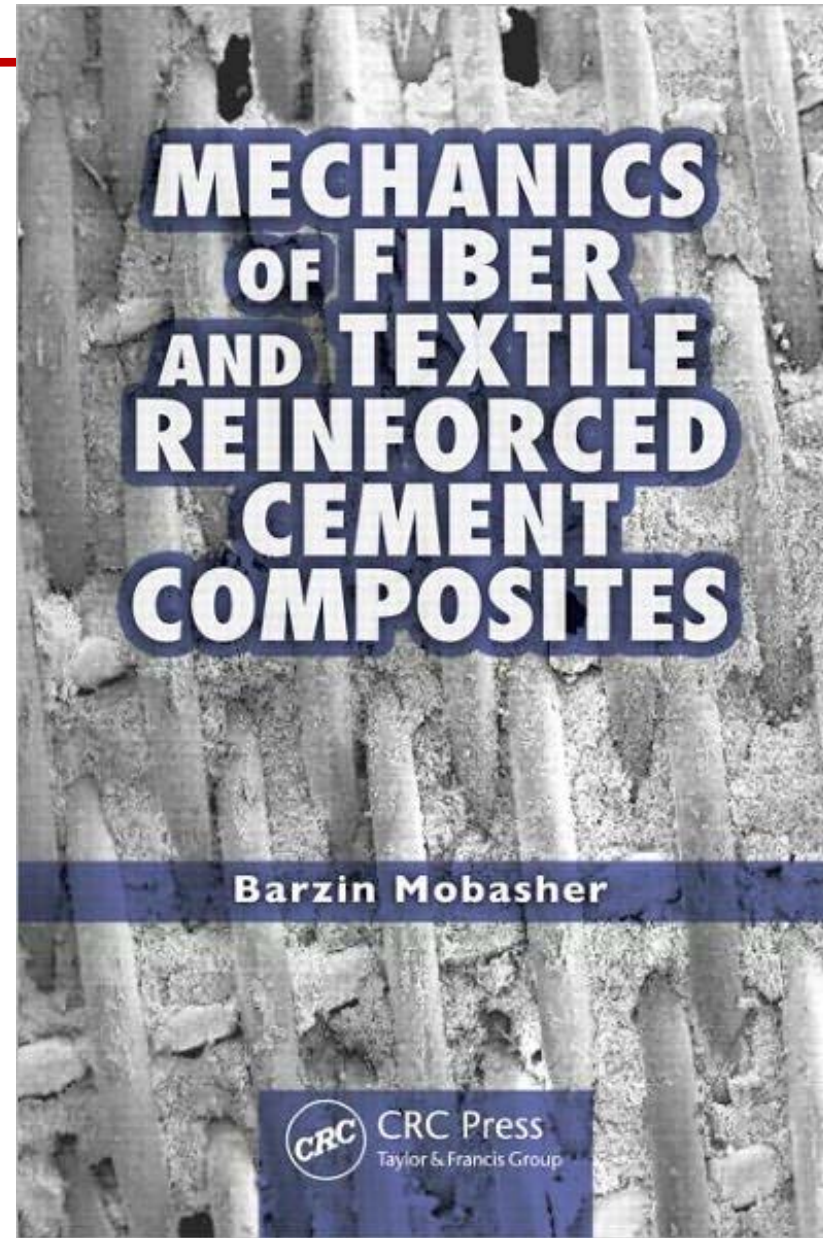


Load Case2:

- 1.4 Self weight +
1.7 Earth pressure +
1.7 Uniform pressure due to surcharge
- Moment in short span direction SM1

Mechanics of Fiber and Textile Reinforced Cement Composites

- CRC press, 2011
- E-mail: barzin@asu.edu
- <http://enpub.fulton.asu.edu/cement/>



Conclusions

- New technologies and directions are clearly available in our future progression in Construction industry.
- Use of materials science based and mechanics based approaches to obtain better ways to characterize, model, analyze, and design.
- Better train, communicate, and follow through
 - Design of Materials and Structures for novel construction.
- Address specifications, quality, and long term performance.
- Fiber Reinforced Concrete can be effectively used as a technology enabler for our alternative energy generation, use and infrastructure development